

A Multiband Photometric Re-assessment of Proposed Planet-driven Stellar Pulsations in HAT-P-2

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I am an undergraduate researcher in the Department of Physics at Penn State working with Dr. John Zanzetti on a multiwavelength reassessment of claimed planet-induced stellar pulsations in the eccentric hot Jupiter system HAT-P-2. Hot Jupiters on highly eccentric orbits probe the same tidal physics seen in “heartbeat” binaries, where strong, time-variable tidal forcing at periastron can excite stellar oscillations at harmonics of the orbital frequency. In HAT-P-2, separate analyses of Spitzer and HST light curves have reported candidate pulsations at integer multiples of the planet’s orbital frequency, but current stellar and tidal models cannot reproduce the reported amplitudes, raising the possibility that these signals are instrumental in origin rather than genuine planet-driven modes. My project uses archival TESS, HST, and Spitzer/IRAC photometry, together with HIRES and HARPS-N radial velocities, to test whether these claimed pulsations are coherent and consistent across wavelength, or instead associated with instrument-specific systematics.

Over the past academic year I have built and validated the core analysis pipeline for the optical (TESS) and near-UV datasets (HST), and I am now extending to the mid-IR regime (Spitzer). Using joint transit and phase-curve modeling (batman + emcee), I fit and subtract the planetary signal from the TESS and HST light curves to isolate residual stellar variability, then apply generalized Lomb–Scargle periodograms and injection-recovery tests to search for power at the reported orbital harmonics. In both TESS and HST, I find no significant peaks at the claimed frequencies above conservative false-alarm thresholds, and injection–recovery experiments demonstrate that the pipeline would have detected pulsations with the reported amplitudes if they were present. These results already strongly suggest that the previously reported signals are not robustly present in the optical and near-UV bands.

PSSC funds would support a focused summer research period to complete the remaining analysis, integrate the multiwavelength data sets, and prepare the work for publication and conference presentation. Scientifically, the next steps are (1) to carry out the same homogeneous transit, phase-curve, and residual analysis on the Spitzer/IRAC light curves, providing the critical infrared comparison to the existing TESS and HST results, and (2) to finalize a consistent Keplerian solution using the HIRES and HARPS-N radial velocities, tying any candidate pulsation frequencies precisely to the orbital ephemeris. Together, these steps will yield a decisive, three-band test of whether HAT-P-2 truly hosts planet-induced heartbeat-like oscillations or whether earlier detections were driven by instrument-specific artifacts.

Because this project is entirely computational and relies solely on archival data, it is well suited to a remote summer research format. However, even without lab or software expenses, the project still requires sustained time and attention to complete the remaining analysis at a publication-ready level. PSSC support would provide financial support during the summer so that I can devote consistent effort to the project rather than splitting my time with unrelated paid work. That protected time would allow me to finish the Spitzer and radial-velocity analysis, consolidate the multiwavelength results, and prepare a first-author manuscript for submission. In this way, the award would directly support the completion and dissemination of an research project that addresses an open question in planetary system tidal dynamics.

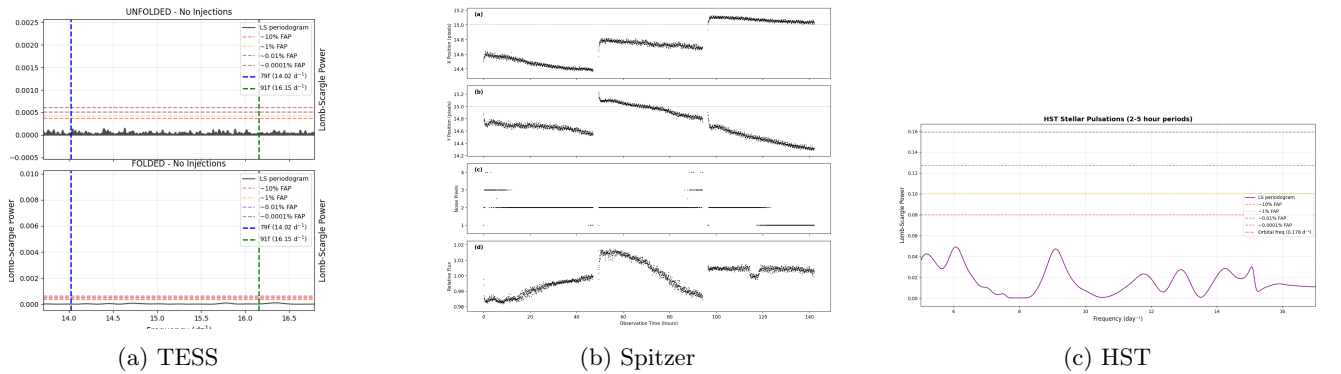


Figure 1: (a) TESS Lomb-Scargle periodograms without injected signals, showing no significant pulsation power above the false-positive probability lines; (b) raw flux and centroid-position plots for the 4.5 μm Spitzer channel after image processing; and (c) the HST Lomb-Scargle periodogram without injected signals, again showing no significant power above the false-positive probability lines.